CELESTIAL MECHANICS

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CHAPTER OVERVIEW

1: Numerical Methods

This chapter is not intended as a comprehensive course in numerical methods. Rather it deals, and only in a rather basic way, with the very common problems of numerical integration and the solution of simple (and not so simple!) Equations. Specialist astronomers today can generate most of the planetary tables for themselves; but those who are not so specialized still have a need to look up data in tables such as The Astronomical Almanac, and I have therefore added a brief section on interpolation, which I hope may be useful. While any of these topics could be greatly expanded, this section should be useful for many everyday computational purposes.

Thumbnail: Comparison between 2-point Gaussian and trapezoidal quadrature. The blue line is the polynomial , whose integral in [−1, 1] is 2/3. The trapezoidal rule returns the integral of the orange dashed line. The 2-point Gaussian quadrature rule returns the integral of the black dashed curve. Such a result is exact, since the green region has the same area as the red regions. (CC BY-Sa 4.0; Paolostar).

1.1: Introduction to Numerical Methods

I believe that, when I was a young student, I had some vague naive belief that every Equation had as its solution an explicit algebraic formula, and every function to be integrated had a similar explicit analytical function for the answer. It came as quite an eye-opener to me when I began to realize that this was far from the case. There are many mathematical operations for which there is no explicit formula, and yet more for which numerical solutions are either easier, or quicker or more convenient than algebraic solutions. I also remember being impressed as a student with the seemingly endless number of "special functions" whose properties were listed in textbooks and which I feared I would have to memorize and master. Of course, we now have computers, and over the years I have come to realize that it is often easier to generate numerical solutions to problems rather than try to express them in terms of obscure special functions with which few people are honestly familiar. Now, far from believing that every problem has an explicit algebraic solution, I suspect that algebraic solutions to problems may be a minority, and numerical solutions to many problems are the norm.

This chapter is not intended as a comprehensive course in numerical methods. Rather it deals, and only in a rather basic way, with the very common problems of numerical integration and the solution of simple (and not so simple!) Equations. Specialist astronomers today can generate most of the planetary tables for themselves; but those who are not so specialized still have a need to look up data in tables such as The Astronomical Almanac, and I have therefore added a brief section on interpolation, which I hope may be useful. While any of these topics could be greatly expanded, this section should be useful for many everyday computational purposes.

I do not deal in this introductory chapter with the huge subject of differential Equations. These need a book in themselves. Nevertheless, there is an example I remember from student days that has stuck in my mind ever since. In those days, calculations were done by hand-operated mechanical calculators, one of which I still fondly possess, and speed and efficiency, as well as accuracy, were a prime concern - as indeed they still are today in an era of electronic computers of astonishing speed. The problem was this: Given the differential Equation

$$\frac{dy}{dx} = \frac{x+y}{x-y} \tag{1.1.1}$$

with initial conditions $y = 0$ when $x = 1$, tabulate y as a function of x. It happens that the differential Equation can readily be solved analytically:

$$\ln(x^2 + y^2) = 2\tan^{-1}(y/x) \tag{1.1.2}$$

Yet it is far quicker and easier to tabulate y as a function of x using numerical techniques directly from the original differential Equation 1.1.1 than from its analytical solution 1.1.2.

1.2: Numerical Integration

There are many occasions when one may wish to integrate an expression numerically rather than analytically. Sometimes one cannot find an analytical expression for an integral, or, if one can, it is so complicated that it is just as quick to integrate numerically as it is to tabulate the analytical expression. Or one may have a table of numbers to integrate rather than an analytical Equation. Many computers and programmable calculators have internal routines for integration, which one can call upon (at risk) without having any idea how they work. It is assumed that the reader of this chapter, however, wants to be able to carry out a numerical integration without calling upon an existing routine that has been written by somebody else.

There are many different methods of numerical integration, but the one known as Simpson's Rule is easy to program, rapid to perform and usually very accurate. (Thomas Simpson, 1710 - 1761, was an English mathematician, author of *A New Treatise on Fluxions*.)

Suppose we have a function $y(x)$ that we wish to integrate between two limits. We calculate the value of the function at the two limits and halfway between, so we now know three points on the curve. We then fit a parabola to these three points and find the area under that.

In the figure I.1, $y(x)$ is the function we wish to integrate between the limits $x_2 - \delta x$ and $x_2 + \delta x$. In other words, we wish to calculate the area under the curve. y_1, y_2 and y_3 are the values of

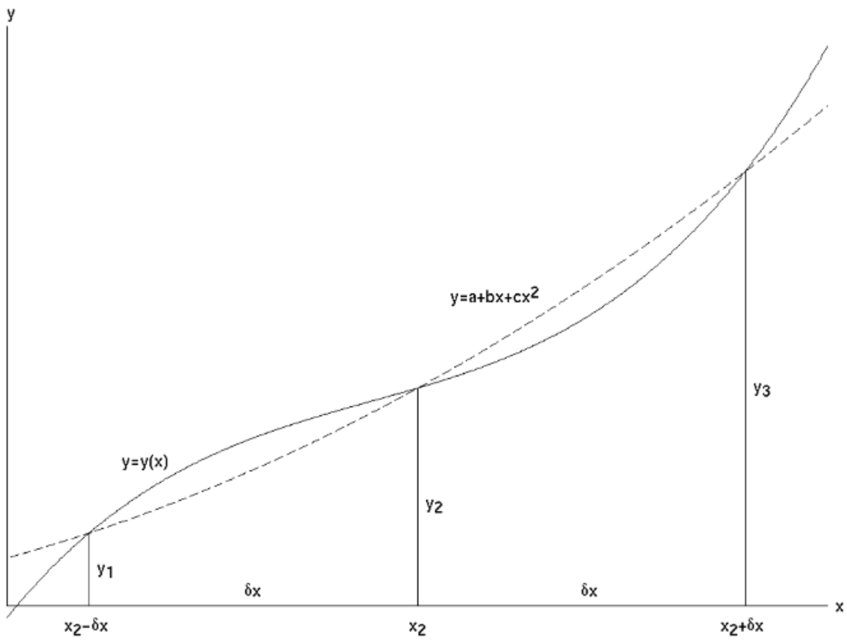


FIGURE I.1 Simpson's Rule gives us the area under the parabola (dashed curve) that passes through three points on the curve $y = y(x)$. This is approximately equal to the area under $y = y(x)$.

the function at $x_2 - \delta_x,\ x_2$ and $x_2 + \delta_x$, and $y = a + bx + cx^2$ is the parabola passing through the points $(x_2 - \delta x, y_1)$, (x_2, y_2) and $(x_2 + \delta x, y_3)$.

If the parabola is to pass through these three points, we must have

$$y_1 = a + b(x_2 - \delta x) + c(x_2 - \delta x)^2 \tag{1.2.1}$$

$$y_2 = a + bx + cx^2 \tag{1.2.2}$$

$$y_3 = a + b(x_2 + \delta x) + c(x_2 + \delta x)^2 \tag{1.2.3}$$

We can solve these Equations to find the values of a, b and c. These are

$$a = y_2 - \frac{x_2(y_3 - y_1)}{2\delta x} + \frac{x_2^2(y_3 - 2y_2 + y_1)}{2(\delta x)^2} \tag{1.2.4}$$

$$y_3 - y_1 \qquad x_2(y_3 - 2y_2 + y_1)$$

$$b=\frac{y_3-y_1}{2\delta x}-\frac{x_2(y_3-2y_2+y_1)}{(\delta x)^2} \tag{1.2.5}$$

$$c=\frac{y_3-2y_2+y_1}{2(\delta x)^2} \tag{1.2.6}$$

Now the area under the parabola (which is taken to be approximately the area under $y(x)$) is

$$\int_{x_2-\delta x}^{x_2+\delta x}\left(a+bx+cx^2\right)dx=2\left[a+bx_2+cx_2^2+\frac{1}{3}c(\delta x)^2\right]\delta x \tag{1.2.7}$$

On substituting the values of a, b and c, we obtain for the area under the parabola

$$\frac{1}{3}(y_1+4y_2+y_3)\delta x \tag{1.2.8}$$

and this is the formula known as Simpson's Rule.

For an example, let us evaluate $\int_0^{\pi/2}\sin x dx$.

We shall evaluate the function at the lower and upper limits and halfway between. Thus

$$\begin{aligned} &x=0, && y=0 \\ &x=\pi/4, && y=1/\sqrt{2} \\ &x=\pi/2, && y=1 \end{aligned}$$

The interval between consecutive values of x is $\delta x=\pi/4$.

Hence Simpson's Rule gives for the area

$$\frac{1}{3}\left(0+\frac{4}{\sqrt{2}}+1\right)\frac{\pi}{4}$$

which, to three significant figures, is 1.00. Graphs of $\sin x$ and $a+bx+cx^2$ are shown in figure $\mathrm{I.2a}$. The values of a, b and c, obtained from the formulas above, are

$$a=0, \quad b=\frac{\sqrt{32}-2}{\pi}, \quad c=\frac{8-\sqrt{128}}{\pi^2}$$

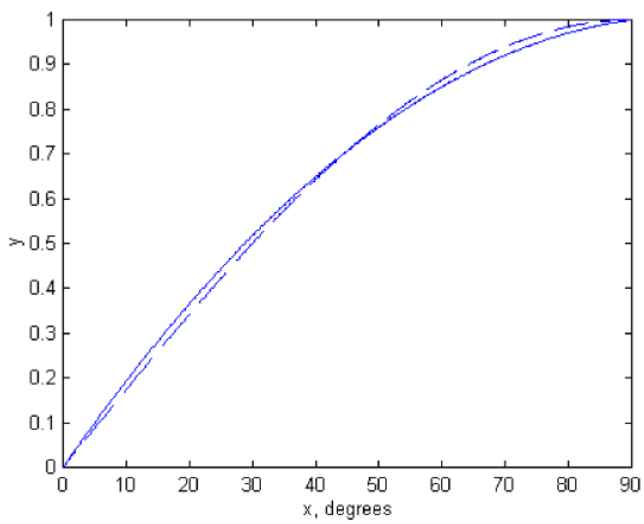


FIGURE I.2a

The result we have just obtained is quite spectacular, and we are not always so lucky. Not all functions can be approximated so well by a parabola. But of course the interval $\delta x=\pi/4$ was ridiculously coarse. In practice we subdivide the interval into numerous very small intervals. For example, consider the integral

$$\int_0^{\pi/4}\cos^{\frac{3}{2}}2x\sin x\,dx.$$

Let us subdivide the interval 0 to $\pi/4$ into ten intervals of width $\pi/40$ each. We shall evaluate the function at the end points and the nine points between, thus:

x	$\cos^{\frac{3}{2}} x \sin x dx$
0	$y_1 = 0.000\ 000\ 000$
$\pi/40$	$y_2 = 0.077\ 014\ 622$
$2\pi/40$	$y_3 = 0.145\ 091\ 486$
$3\pi/40$	$y_4 = 0.196\ 339\ 002$
$4\pi/40$	$y_5 = 0.224\ 863\ 430$
$5\pi/40$	$y_6 = 0.227\ 544\ 930$
$6\pi/40$	$y_7 = 0.204\ 585\ 473$
$7\pi/40$	$y_8 = 0.159\ 828\ 877$
$8\pi/40$	$y_9 = 0.100\ 969\ 971$
$9\pi/40$	$y_{10} = 0.040\ 183\ 066$
$10\pi/40$	$y_{11} = 0.000\ 000\ 000$

The integral from 0 to $2\pi/40$ is $\frac{1}{3}(y_1 + 4y_2 + y_3)\delta x$, δx being the interval $\pi/40$. The integral from $3\pi/40$ to $4\pi/40$ is $\frac{1}{3}(y_3 + 4y_4 + y_5)\delta x$. And so on, until we reach the integral from $8\pi/40$ to $10\pi/40$. When we add all of these up, we obtain for the integral from 0 to $\pi/4$,

$$\frac{1}{3}\left(y_1 + 4y_2 + 2y_3 + 4y_4 + 2y_5 + \ldots\ldots + 4y_{10} + y_{11}\right)\delta x$$

$$= \frac{1}{3}[y_1 + y_{11} + (y_2 + y_4 + y_6 + y_8 + y_{10}) + 2(y_3 + y_5 + y_7 + y_9)]\delta x,$$

which comes to $0.108\ 768\ 816$.

We see that the calculation is rather quick, and it is easily programmable (try it!). But how good is the answer? Is it good to three significant figures? Four? Five?

Since it is fairly easy to program the procedure for a computer, my practice is to subdivide the interval successively into 10, 100, 1000 subintervals, and see whether the result converges. In the present example, with N subintervals, I found the following results:

N	integral
10	0.108 768 816
100	0.108 709 621
1000	0.108 709 466
10000	0.108 709 465

This shows that, even with a course division into ten intervals, a fairly good result is obtained, but you do have to work for more significant figures. I was using a mainframe computer when I did the calculation with 10000 intervals, and the answer was displayed on my screen in what I would estimate was about one fifth of a second.

There are two more lessons to be learned from this example. One is that sometimes a change of variable will make things very much faster. For example, if one makes one of the (fairly obvious?) trial substitutions $y = \cos x$, $y = \cos 2x$ or $y^2 = \cos 2x$, the integral becomes

$$\int_{1/\sqrt{2}}^{1} (2y^2 - 1)^{3/2} dy, \quad \int_0^1 \sqrt{\frac{y^3}{8(1+y)}} dy \quad \text{or} \quad \int_0^1 \frac{y^4}{\sqrt{2(1+y^2)}} dy.$$

Not only is it very much faster to calculate any of these integrands than the original trigonometric expression, but I found the answer $0.108\ 709\ 465$ by Simpson's rule on the third of these with only 100 intervals rather than $10,000$, the answer appearing on the screen apparently instantaneously. (The first two required a few more intervals.)

To gain about one fifth of a second may appear to be of small moment, but in truth the computation went faster by a factor of several hundred. One sometimes hears of very large computations involving massive amounts of data requiring overnight computer runs of eight hours or so. If the programming speed and efficiency could be increased by a factor of a few hundred, as in this example, the entire computation could be completed in less than a minute.

The other lesson to be learned is that the integral does, after all, have an explicit algebraic form. You should try to find it, not only for integration practice, but to convince yourself that there are indeed occasions when a numerical solution can be found faster than an analytic one! The answer, by the way, is $\frac{\sqrt{18}\ln(1+\sqrt{2})-2}{16}$.

You might now like to perform the following integration numerically, either by hand calculator or by computer.

$$\int_0^2 \frac{x^2 dx}{\sqrt{2-x}}$$

At first glance, this may look like just another routine exercise, but you will very soon find a small difficulty and wonder what to do about it. The difficulty is that, at the upper limit of integration, the integrand becomes infinite. This sort of difficulty, which is not infrequent, can often be overcome by means of a change of variable. For example, let $x = 2\sin^2\theta$, and the integral becomes

$$8\sqrt{2}\int_0^{\pi/2} \sin^5\theta d\theta$$

and the difficulty has gone. The reader should try to integrate this numerically by Simpson's rule, though it may also be noted that it has an exact analytic answer, namely $\sqrt{8192}/15$.

Here is another example. It can be shown that the period of oscillation of a simple pendulum of length l swinging through 90° on either side of the vertical is

$$P = \sqrt{\frac{8l}{g}}\int_0^{\pi/2} \sqrt{\sec\theta} d\theta.$$

As in the previous example, the integrand becomes infinite at the upper limit. I leave it to the reader to find a suitable change of variable such that the integrand is finite at both limits, and then to integrate it numerically. (If you give up, see Section 1.13.) Unlike the last example, this one has no simple analytic solution in terms of elementary functions. It can be written in terms of special functions (elliptic integrals) but they have to be evaluated numerically in any case, so that is of little help. I make the answer

$$P = 2.3607\pi\sqrt{\frac{l}{g}}.$$

For another example, consider

$$\int_0^\infty \frac{dx}{x^5\left(e^{1/x}-1\right)}$$

This integral occurs in the theory of blackbody radiation. To help you to visualize the integrand, it and its first derivative are zero at $x=0$ and $x=\infty$ and it reaches a maximum value of 21.201435 at $x=0.201405$. The difficulty this time is the infinite upper limit. But, as in the previous two examples, we can overcome the difficulty by making a change of variable. For example, if we let $x = \tan\theta$, the integral becomes

$$\int_0^{\pi/2} \frac{c^3\left(c^2+1\right) d\theta}{e^c-1}, \text{where } c = \cot\theta = 1/x.$$

The integrand is zero at both limits and is easily calculable between, and the value of the integral can now be calculated by Simpson's rule in a straightforward way. It also has an exact analytic solution, namely $\pi^4/15$, though it is hard to say whether it is easier to arrive at this by analysis or by numerical integration.

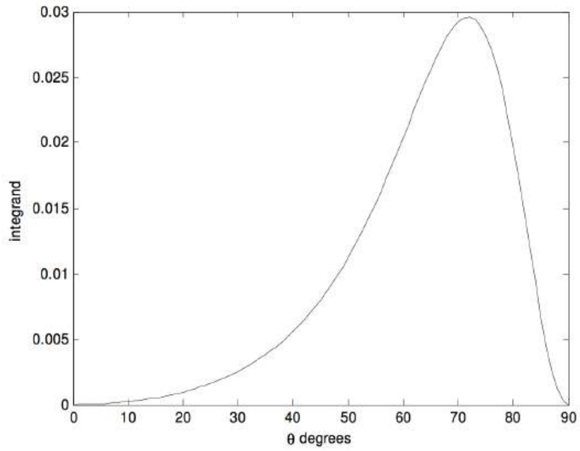
Here's another:

$$\int_0^\infty \frac{x^2 dx}{(x^2+9)(x^2+4)^2}$$

The immediate difficulty is the infinite upper limit, but that is easily dealt with by making a change of variable: $x=\tan\theta$. The integral then becomes

$$\int_{\theta=0}^{\pi/2} \frac{t(t+1) d\theta}{(t+9)(t+4)^2}$$

in which $t = \tan^2 \theta$. The upper limit is now finite, and the integrand is easy to compute - except, perhaps, at the upper limit. However, after some initial hesitation the reader will probably agree that the integrand is zero at the upper limit. The integrand looks like this:



It reaches a maximum of $0.029\ 5917$ at $\theta = 71^\circ.789\ 962$. Simpson's rule easily gave me an answer of $0.015\ 708$. The integral has an analytic solution (try it) of $\pi/200$.

There are, of course, methods of numerical integration other than Simpson's rule. I describe one here without proof. I call it "seven-point integration". It may seem complicated, but once you have successfully programmed it for a computer, you can forget the details, and it is often even faster and more accurate than Simpson's rule. You evaluate the function at $6n + 1$ points, where n is an integer, so that there are $6n$ intervals. If, for example, $n = 4$, you evaluate the function at 25 points, including the lower and upper limits of integration. The integral is then:

$$\int_a^b f(x)dx = 0.3 \times (\Sigma_1 + 2\Sigma_2 + 5\Sigma_3 + 6\Sigma_4)\delta x, \tag{1.2.9}$$

where δx is the size of the interval, and

$$\Sigma_1 = f_1 + f_3 + f_5 + f_9 + f_{11} + f_{15} + f_{17} + f_{21} + f_{23} + f_{25}, \tag{1.2.10}$$

$$\Sigma_2 = f_7 + f_{13} + f_{19}, \tag{1.2.11}$$

$$\Sigma_3 = f_2 + f_6 + f_8 + f_{12} + f_{14} + f_{18} + f_{20} + f_{24} \tag{1.2.12}$$

and

$$\Sigma_4 = f_4 + f_{10} + f_{16} + f_{22}. \tag{1.2.13}$$

Here, of course, $f_1 = f(a)$ and $f_{25} = f(b)$. You can try this on the functions we have already integrated by Simpson's rule, and see whether it is faster.

Let us try one last integration before moving to the next section. Let us try

$$\int_0^{10} \frac{1}{1 + 8x^3} dx.$$

This can easily (!) be integrated analytically, and you might like to show that it is

$$\frac{1}{12} \ln \frac{147}{127} + \frac{1}{\sqrt{12}} \tan^{-1} \sqrt{507} + \frac{\pi}{\sqrt{432}} = 0.6039748.$$

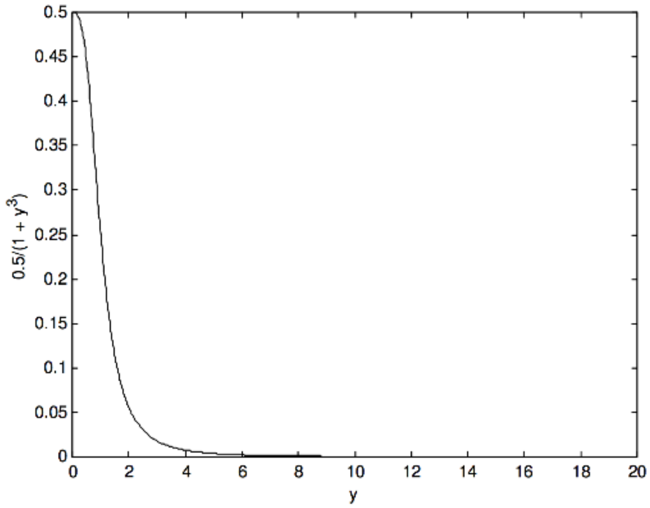
However, our purpose in this section is to learn some skills of numerical integration. Using Simpson's rule, I obtained the above answer to seven decimal places with 544 intervals. With seven-point integration, however, I used only 162 intervals to achieve the same precision, a reduction of 70. Either way, the calculation on a fast computer was almost instantaneous. However, had it been a really lengthy integration, the greater efficiency of the seven point integration might have saved hours. It is also worth noting that $x \times x \times x$ is faster to compute than x^3. Also, if we make the substitution y = 2x, the integral becomes

$$\int^{20} \quad 0.5$$

$$\int_0 \frac{\quad}{1+y^3}dy.$$

This reduces the number of multiplications to be done from 489 to 326 – i.e. a further reduction of one third. But we have still not done the best we could do. Let us look at the function $\frac{0.5}{1+y^3}$, in figure I.2b:

FIGURE I2b



We see that beyond $y=6$, our efforts have been largely wasted. We don't need such fine intervals of integration. I find that I can obtain the same level of precision − i.e. an answer of 0.6039748 − using 48 intervals from $y=0$ to 6 and 24 intervals from $y=6$ to 20. Thus, by various means we have reduced the number of times that the function had to be evaluated from our original 545 to 72, as well as reducing the number of multiplications each time by a third, a reduction of computing time by 91. This last example shows that it often advantageous to use fine intervals of integration only when the function is rapidly changing (i.e. has a large slope), and to revert to coarser intervals where the function is changing only slowly.

The *Gaussian quadrature* method of numerical integration is described in Sections 1.15 and 1.16.

1.3: Quadratic Equations

Any reader of this book will know that the solutions to the quadratic Equation

$$ax^2+bx+c=0 \tag{1.3.1}$$

are

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a} \tag{1.3.2}$$

and will have no difficulty in finding that the solutions to

$$2.9x^2-4.7x+1.7=0$$

are

$$x=1.0758 \text{ or } 0.5449.$$

We are now going to look, largely for fun, at two alternative iterative numerical methods of solving a quadratic Equation. One of them will turn out not to be very good, but the second will turn out to be sufficiently good to merit our serious attention.

In the first method, we re-write the quadratic Equation in the form

$$x=\frac{-\left(ax^2+c\right)}{b}$$

We guess a value for one of the solutions, put the guess in the right hand side, and hence calculate a new value for x. We continue iterating like this until the solution converges.

For example, let us guess that a solution to the Equation $2.9x^2-4.7x+1.7=0$ is $x=0.55$. Successive iterations produce the values

$$\begin{array}{ll}
0.54835 & 0.54501\\
0.54723 & 0.54498\\
0.54648 & 0.54496\\
0.54597 & 0.54495\\
0.54562 & 0.54494\\
0.54539 & 0.54493\\
0.54524 & 0.54493\\
0.54513 & 0.54494\\
0.54506 & 0.54492
\end{array}$$

We did eventually arrive at the correct answer, but it was very slow indeed even though our first guess was so close to the correct answer that we would not have been likely to make such a good first guess accidentally.

Let us try to obtain the second solution, and we shall try a first guess of 1.10, which again is such a good first guess that we would not be likely to arrive at it accidentally. Successive iterations result in

$$\begin{array}{l}
1.10830\\
1.11960\\
1.13515
\end{array}$$

and we are getting further and further from the correct answer!

Let us try a better first guess of 1.05. This time, successive iterations result in

$$\begin{array}{l}
1.04197\\
1.03160\\
1.01834
\end{array}$$

Again, we are getting further and further from the solution.

No more need be said to convince the reader that this is not a good method, so let us try something a little different.

We start with

$$ax^2 + bx = -c \tag{1.3.3}$$

Add ax^2 to each side:

$$2ax^2 + bx = ax^2 - c \tag{1.3.4}$$

or

$$(2ax + b)x = ax^2 - c \tag{1.3.5}$$

Solve for x:

$$x = \frac{ax^2 - c}{2ax + b} \tag{1.3.6}$$

This is just the original Equation written in a slightly rearranged form. Now let us make a guess for x, and iterate as before. This time, however, instead of making a guess so good that we are unlikely to have stumbled upon it, let us make a very stupid first guess, for example $x = 0$. Successive iterations then proceed as follows.

$$\begin{array}{c} 0.00000 \\ 0.36170 \\ 0.51751 \\ 0.54261 \\ 0.54491 \\ 0.54492 \end{array}$$

and the solution converged rapidly in spite of the exceptional stupidity of our first guess. The reader should now try another very stupid first guess to try to arrive at the second solution. I tried $x = 100$, which is very stupid indeed, but I found convergence to the solution 1.0758 after just a few iterations.

Even although we already know how to solve a quadratic Equation, there is something intriguing about this. What was the motivation for adding ax^2 to each side of the Equation, and why did the resulting minor rearrangement lead to rapid convergence from a stupid first guess, whereas a simple direct iteration either converged extremely slowly from an impossibly good first guess or did not converge at all?

1.4: The Solution of f(x) = 0

The title of this section is intended to be eye-catching. Some Equations are easy to solve; others seem to be more difficult. In this section, we are going to try to solve any Equation at all of the form $f(x) = 0$ (which covers just about everything!) and we shall in most cases succeed with ease.

Figure I.3 shows a graph of the Equation $y = f(x)$. We have to find the value (or perhaps values) of x such that $f(x) = 0$.

We guess that the answer might be x_g, for example. We calculate $f(x_g)$. It won't be zero, because our guess is wrong. The figure shows our guess x_g, the correct value x, and $f(x_g)$. The tangent of the angle θ is the derivative $f'(x)$, but we cannot calculate the derivative there because we do not yet know x. However, we can calculate $f'(x_g)$, which is close. In any case $\tan\theta$, or $f'(x_g)$, is approximately equal to $f(x_g)/(x_g - x)$, so that

$$x \approx x_g - \frac{f(x_g)}{f'(x_g)} \tag{1.4.1}$$

will be much closer to the true value than our original guess was. We use the new value as our next guess, and keep on iterating until

$$\left|\frac{x_g - x}{x_g}\right|$$

is less than whatever precision we desire. The method is usually extraordinarily fast, even for a wildly inaccurate first guess. The method is known as **Newton-Raphson iteration**. There are some cases where the method will not converge, and stress is often placed on these exceptional cases in mathematical courses, giving the impression that the Newton-Raphson process is of limited applicability. These exceptional cases are, however, often artificially concocted in order to illustrate the exceptions (we do indeed cite some below), and in practice Newton-Raphson is usually the method of choice.

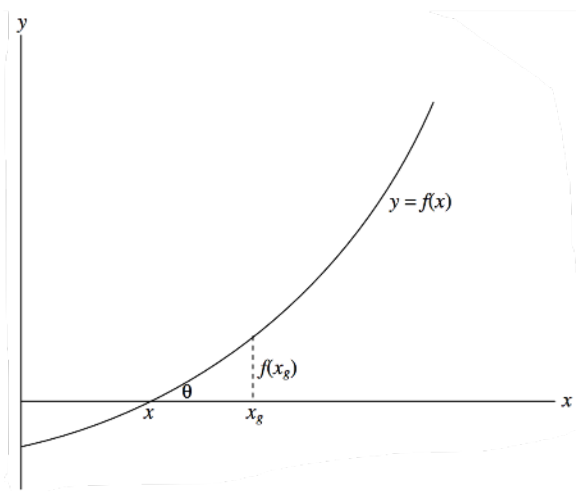


FIGURE I.3

I shall often drop the clumsy subscript g, and shall write the Newton-Raphson scheme as

$$x = x - f(x)/f'(x), \tag{1.4.2}$$

meaning "start with some value of x, calculate the right hand side, and use the result as a new value of x". It may be objected that this is a misuse of the $=$ symbol, and that the above is not really an "Equation", since x cannot equal x minus something. However, when the correct solution for x has been found, it will satisfy $f(x) = 0$, and the above is indeed a perfectly good Equation and a valid use of the $=$ symbol.

Example 1.4.1

Solve the Equation $1/x = \ln x$

We have $f = 1/x - \ln x = 0$

And $f' = -(1+x)/x^2$,

from which $x - f/f'$ becomes, after some simplification,

$$\frac{x[2 + x(1 - \ln x)]}{1 + x},$$

so that the Newton-Raphson iteration is

$$x = \frac{x[2 + x(1 - \ln x)]}{1 + x}.$$

There remains the question as to what should be the first guess. We know (or should know!) that $\ln 1 = 0$ and $\ln 2 = 0.6931$, so the answer must be somewhere between 1 and 2. If we try $x = 1.5$, successive iterations are

$$\begin{array}{c} 1.735\ 081\ 403 \\ 1.762\ 915\ 391 \\ 1.763\ 222\ 798 \\ 1.763\ 222\ 834 \\ 1.763\ 222\ 835 \end{array}$$

This converged quickly from a fairly good first guess of 1.5. Very often the Newton-Raphson iteration will converge, even rapidly, from a very stupid first guess, but in this particular example there are limits to stupidity, and the reader might like to prove that, in order to achieve convergence, the first guess must be in the range

$$0 < x < 4.319\ 136\ 566$$

Example $1.4.2$

Solve the unlikely Equation $\sin x = \ln x$

We have $f = \sin x - \ln x$ and $f' = \cos x - 1/x$,

and after some simplification the Newton-Raphson iteration becomes

$$x = x\left[1 + \frac{\ln x - \sin x}{x \cos x - 1}\right].$$

Graphs of $\sin x$ and $\ln x$ will provide a first guess, but in lieu of that and without having much idea of what the answer might be, we could try a fairly stupid $x = 1$. Subsequent iterations produce

$$\begin{array}{c} 2.830\ 487\ 722 \\ 2.267\ 902\ 211 \\ 2.219\ 744\ 452 \\ 2.219\ 107\ 263 \\ 2.219\ 107\ 149 \\ 2.219\ 107\ 149 \end{array}$$

Example $1.4.3$

Solve the Equation $x^2 = a$ (A new way of finding square roots!)

$$f = x^2 - a, \quad f' = 2x.$$

After a little simplification, the Newton-Raphson process becomes

$$x = \frac{x^2 + a}{2x}.$$

For example, what is the square root of 10? Guess 3. Subsequent iterations are

$$\begin{array}{c} 3.166\ 666\ 667 \\ 3.162\ 280\ 702 \\ 3.162\ 277\ 661 \end{array}$$

$$3.162\ 277\ 661$$

Example 1.4.4

Solve the Equation $ax^2 + bx + c = 0$ (A new way of solving quadratic Equations!)

$$f = ax^2 + bx + c = 0,$$

$$f' = 2ax + b.$$

Newton-Raphson:

$$x = x - \frac{ax^2 + bx + c}{2ax + b},$$

which becomes, after simplification,

$$x = \frac{ax^2 - c}{2ax + b}.$$

This is just the iteration given in the previous section, on the solution of quadratic Equations, and it shows why the previous method converged so rapidly and also how I really arrived at the Equation (which was via the Newton-Raphson process, and not by arbitrarily adding ax^2 to both sides!)

1.5: The Solution of Polynomial Equations

The Newton-Raphson method is very suitable for the solution of polynomial Equations, for example for the solution of a quintic Equation:

$$a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 = 0. \tag{1.5.1}$$

Before illustrating the method, it should be pointed out that, even though it may look inelegant in print, in order to evaluate a polynomial expression numerically it is far easier and quicker to nest the parentheses and write the polynomial in the form

$$a_0 + x(a_1 + x(a_2 + x(a_3 + x(a_4 + xa_5)))). \tag{1.5.2}$$

Working from the inside out, we see that the process is a multiplication followed by an addition, repeated over and over. This is very easy whether the calculation is done by computer, by calculator, or in one's head.

For example, evaluate the following expression in your head, for $x = 4$:

$$2 - 7x + 2x^2 - 8x^3 - 2x^4 + 3x^5.$$

You couldn't? But now evaluate the following expression in your head for $x = 4$ and see how (relatively) easy it is:

$$2 + x(-7 + x(2 + x(-8 + x(-2 + 3x)))).$$

Fortran

As an example of how efficient the nested parentheses are in a computer program, here is a FORTRAN program for evaluating a fifth degree polynomial. It is assumed that the value of x has been defined in a FORTRAN variable called X, and that the six coefficients $a_0, a_1, \ldots a_5$ have been stored in a vector as $\mathrm{A}(1),\ \mathrm{A}(2), \ldots\ \mathrm{A}(6)$.

$$\begin{array}{ll} & \mathrm{Y = 0.} \\ & \mathrm{DO1I = 1,5} \\ 1 & \mathrm{Y = (Y + A(7-I))^*X} \\ & \mathrm{Y = Y + A(1)} \end{array}$$

The calculation is finished!

We return now to the solution of

$$f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 = 0. \tag{1.5.3}$$

We have

$$f'(x) = a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + 5a_5x^4. \tag{1.5.4}$$

Now

$$x = x - f/f', \tag{1.5.5}$$

and after simplification,

$$x = \frac{-a_0 + x^2(a_2 + x(2a_3 + x(3a_4 + 4a_5x)))}{a_1 + x(2a_2 + x(3a_3 + x(4a_4 + 5a_5x)))}, \tag{1.5.6}$$

which is now ready for numerical iteration.

For example, let us solve

$$205 + 111x + 4x^2 - 31x^3 - 10x^4 + 3x^5 = 0 \tag{1.5.7}$$

A reasonable first guess could be obtained by drawing a graph of this function to see where it crosses the x-axis, but, truth to tell, the Newton-Raphson process usually works so well that one need spend little time on a first guess; just use the first number that comes into your head, for example, $x = 0$. Subsequent iterations then go

Illegal pream-token (-)

A question that remains is: How many solutions are there? The general answer is that an nth degree polynomial Equation has n solutions. This statement needs to be qualified a little. For example, the solutions need not be real. The solutions may be imaginary, as they are, for example, in the Equation

$$1+x^2=0 \tag{1.5.8}$$

or complex, as they are, for example, in the Equation

$$1+x+x^2=0. \tag{1.5.9}$$

If the solutions are real they may not be distinct. For example, the Equation

$$1-2x+x^2=0 \tag{1.5.10}$$

has two solutions at $x=1$, and the reader may be forgiven for thinking that this somewhat stretches the meaning of "two solutions". However, if one includes complex roots and repeated real roots, it is then always true that an nth degree polynomial has n solutions. The five solutions of the quintic Equation we solved above, for example, are

$$\begin{array}{lcl} 4.947\ 845 & & \\ 2.340\ 216 & & \\ -1.967\ 110 & & \\ -0.993\ 808 & + & 1.418\ 597i \\ -0.993\ 808 & - & 1.418\ 597i \end{array}$$

Can one tell in advance how many real roots a polynomial Equation has? The most certain way to tell is to plot a graph of the polynomial function and see how many times it crosses the x-axis. However, it is possible to a limited extent to determine in advance how many real roots there are. The following "rules" may help. Some will be fairly obvious; others require proof.

The number of real roots of a polynomial of odd degree is odd. Thus a quintic Equation can have one, three or five real roots. Not all of these roots need be distinct, however, so this is of limited help. Nevertheless a polynomial of odd degree always has at least one real root. The number of real roots of an Equation of even degree is even - but the roots need not all be distinct, and the number of real roots could be zero.

An upper limit to the number of real roots can be determined by examining the signs of the coefficients. For example, consider again the Equation

$$205+111x+4x^2-31x^3-10x^4+3x^5=0. \tag{1.5.11}$$

The signs of the coefficients, written in order starting with a_0, are

$$+++--+$$

Run your eye along this list, and count the number of times there is a change of sign. The sign changes twice. This tells us that there are not more than two positive real roots. (If one of the coefficients in a polynomial Equation is zero, i.e. if one of the terms is "missing", this does not count as a change of sign.)

Now change the signs of all coefficients of odd powers of x:

$$+-++--$$

This time there are three changes of sign. This tells us that there are not more than three negative real roots.

In other words, the number of changes of sign in $f(x)$ gives us an upper limit to the number of positive real roots, and the number of changes of sign in $f(-x)$ gives us an upper limit to the number of negative real roots.

One last "rule" is that complex roots occur in conjugate pairs. In our particular example, these rules tell us that there are not more than two positive real roots, and not more than three negative real roots. Since the degree of the polynomial is odd, there is at least one real root, though we cannot tell whether it is positive or negative.

In fact the particular Equation, as we have seen, has two positive real roots, one negative real root, and two conjugate complex roots.

1.6: Failure of the Newton-Raphson Method

This section is written reluctantly, for fear it may give the impression that the Newton-Raphson method frequently fails and is of limited usefulness. This is not the case; in nearly all cases encountered in practice it is very rapid and does not require a particularly good first guess. Nevertheless for completeness it should be pointed out that there are rare occasions when the method either fails or converges rather slowly.

One example is the quintic Equation that we have just encountered:

$$205+111x+4x^2-31x^3-10x^4+5x^5=0 \tag{1.6.1}$$

When we chose $x=0$ as our first guess, we reached a solution fairly quickly. If we had chosen $x=1$, we would not have been so lucky, for the first iteration would have taken us to -281, a very long way from any of the real solutions. Repeated iteration will eventually take us to the correct solution, but only after many iterations. This is not a typical situation, and usually almost any guess will do.

Another example of an Equation that gives some difficulty is

$$x=\tan x, \tag{1.6.2}$$

an Equation that occurs in the theory of single-slit diffraction.

We have

$$f(x)=x-\tan x=0 \tag{1.6.3}$$

and

$$f'(x)=1-\sec^2 x=-\tan^2 x. \tag{1.6.4}$$

The Newton-Raphson process takes the form

$$x=x+\frac{x-\tan x}{\tan^2 x}. \tag{1.6.5}$$

The solution is $x=4.493\ 409$, but in order to achieve this the first guess must be between 4.3 and 4.7. This again is unusual, and in most cases almost any reasonable first guess results in rapid convergence.

The Equation

$$1-4x+6x^2-4x^3+x^4=0 \tag{1.6.6}$$

is an obvious candidate for difficulties. The four identical solutions are $x=1$, but at $x=1$ not only is $f(x)$ zero, but so is $f'(x)$. As the solution $x=1$ is approached, convergence becomes very slow, but eventually the computer or calculator will record an error message as it attempts to divide by the nearly zero $f'(x)$.

I mention just one last example very briefly. When discussing orbits, we shall encounter an Equation known as Kepler's Equation. The Newton-Raphson process almost always solves Kepler's Equation with spectacular speed, even with a very poor first guess. However, there are some very rare occasions. almost never encountered in practice, where the method fails. We shall discuss this Equation in Chapter 9.

1.7: Simultaneous Linear Equations, N = n

Consider the Equations

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + a_{14}x_4 + a_{15}x_5 = b_1 \tag{1.7.1}$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + a_{24}x_4 + a_{25}x_5 = b_2 \tag{1.7.2}$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 + a_{34}x_4 + a_{35}x_5 = b_3 \tag{1.7.3}$$

$$a_{41}x_1 + a_{42}x_2 + a_{43}x_3 + a_{44}x_4 + a_{45}x_5 = b_4 \tag{1.7.4}$$

$$a_{51}x_1 + a_{52}x_2 + a_{53}x_3 + a_{54}x_4 + a_{55}x_5 = b_5 \tag{1.7.5}$$

There are two well-known methods of solving these Equations. One of these is called Cramer's Rule. Let D be the determinant of the coefficients. Let D_i be the determinant obtained by substituting the column vector of the constants $b_1, \; b_2, \; b_3, \; b_4, \; b_5$ for the ith column in D. Then the solutions are

$$x_i = D_i/D \tag{1.7.6}$$

This is an interesting theorem in the theory of determinants. It should be made clear, however, that, when it comes to the practical numerical solution of a set of linear Equations that may be encountered in practice, this is probably the most laborious and longest method ever devised in the history of mathematics.

The second well-known method is to write the Equations in matrix form:

$$\mathbb{A}\mathbf{x} = \mathbf{b} \tag{1.7.7}$$

Here $\mathbb{A}$ is the matrix of the coefficients, $\mathbf{x}$ is the column vector of unknowns, and $\mathbf{b}$ is the column vector of the constants. The solutions are then given by

$$\mathbf{x} = \mathbb{A}^{-1}\mathbf{b}, \tag{1.7.8}$$

where $\mathbb{A}^{-1}$ is the inverse or reciprocal of $\mathbb{A}$. Thus the problem reduces to inverting a matrix. Now inverting a matrix is notoriously labour-intensive, and, while the method is not quite so long as Cramer's Rule, it is still far too long for practical purposes.

How, then, should a system of linear Equations be solved?

Consider the Equations

$$7x - 2y = 24 \tag{1.7.8}$$

$$3x + 9y = 30 \tag{1.7.9}$$

Few would have any hesitation in multiplying the first Equation by 3, the second Equation by 7, and subtracting. This is what we were all taught in our younger days, but few realize that this remains, in spite of knowledge of determinants and matrices, the fastest and most efficient method of solving simultaneous linear Equations. Let us see how it works with a system of several Equations in several unknowns.

Consider the Equations

$$9x_1 - 9x_2 + 8x_3 - 6x_4 + 4x_5 = -9 \tag{1.7.10}$$

$$5x_1 - x_2 + 6x_3 + x_4 + 5x_5 = 58 \tag{1.7.11}$$

$$2x_1 + 4x_2 - 5x_3 - 6x_4 + 7x_5 = -1 \tag{1.7.12}$$

$$2x_1 + 3x_2 - 8x_3 - 5x_4 - 2x_5 = -49 \tag{1.7.13}$$

$$8x_1 - 5x_2 + 7x_3 + x_4 + 5x_5 = 42 \tag{1.7.14}$$

We first eliminate x_1 from the Equations, leaving four Equations in four unknowns. Then we eliminate x_2, leaving three Equations in three unknowns. Then x_3, and then x_4, finally leaving a single Equation in one unknown. The following table shows how it is done.

In columns 2 to 5 are listed the coefficients of x_1, x_2, x_3, x_4 and x_5, and in column 6 are the constant terms on the right hand side of the Equations. Thus columns 2 to 6 of the first five rows are just the original Equations. Column 7 is the sum of the numbers in columns 2 to 6, and this is a most important column. The boldface numbers in column 1 are merely labels.

Lines 6 to 9 show the elimination of x_1. Line 6 shows the elimination of x_1 from lines 1 and 2 by multiplying line 2 by 9 and line 1 by 5 and subtracting. The operation performed is recorded in column 1. In line 7, x_1 is eliminated from Equations 1 and 3 and so on.

$$\begin{array}{lccccccc}
 & x_1 & x_2 & x_3 & x_4 & x_5 & b & \sum \\
\mathbf{1} & 9 & -9 & 8 & -6 & 4 & -9 & -3 \\
\mathbf{2} & 5 & -1 & 6 & 1 & 5 & 58 & 74 \\
\mathbf{3} & 2 & 4 & -5 & -6 & 7 & -1 & 1 \\
\mathbf{4} & 2 & 3 & -8 & -5 & -2 & -49 & -59 \\
\mathbf{5} & 8 & -5 & 7 & 1 & 5 & 42 & 58 \\
 & & & & & & & \\
\mathbf{6} = 9\times\mathbf{2}-5\times\mathbf{1} & & 36 & 14 & 39 & 25 & 567 & 681 \\
\mathbf{7} = 2\times\mathbf{1}-9\times\mathbf{3} & & -54 & 61 & 42 & -55 & -9 & -15 \\
\mathbf{8} = \mathbf{3}-\mathbf{4} & & 1 & 3 & -1 & 9 & 48 & 60 \\
\mathbf{9} = 4\times\mathbf{3}-\mathbf{5} & & 21 & -27 & -25 & 23 & -46 & -54 \\
 & & & & & & & \\
\mathbf{10} = 3\times\mathbf{6}+2\times\mathbf{7} & & & 164 & 201 & -35 & 1\ 683 & 2\ 013 \\
\mathbf{11} = \mathbf{6}-36\times\mathbf{8} & & & -94 & 75 & -299 & -1\ 161 & -1\ 479 \\
\mathbf{12} = 7\times\mathbf{6}-12\times\mathbf{9} & & & 422 & 573 & -101 & 4\ 521 & 5\ 415 \\
 & & & & & & & \\
\mathbf{13} = 47\times\mathbf{10}+82\times\mathbf{11} & & & & 15\ 597 & -26\ 163 & -16\ 101 & -26\ 667 \\
\mathbf{14} = 211\times\mathbf{11}+47\times\mathbf{12} & & & & 42\ 756 & -67\ 836 & -32\ 484 & -57\ 654 \\
 & & & & & & & \\
\mathbf{15} = 5199\times\mathbf{14}-14252\times\mathbf{13} & & & & & 20\ 195\ 712 & 60\ 587\ 136 & 80\ 782\ 848
\end{array} \tag{1.7.15}$$

The purpose of Σ ? This column is of great importance. Whatever operation is performed on the previous columns is also performed on Σ, and Σ must remain the sum of the previous columns. If it does not, then an arithmetic mistake has been made, and it is immediately detected. There is nothing more disheartening to discover at the very end of a calculation that a mistake has been made and that one has no idea where the mistake occurred. Searching for mistakes takes far longer than the original calculation. The Σ-column enables one to detect and correct a mistake as soon as it has been made.

We eventually reach line 15, which is

$$20\ 195\ 712 x_5 = 60\ 587\ 136, \tag{1.7.16}$$

from which

$$x_5 = 3. \tag{1.7.17}$$

x_4 can now easily be found from either or both of lines 13 and 14, x_3 can be found from any or all of lines 10, 11 and 12, and so on. When the calculation is complete, the answers should be checked by substitution in the original Equations (or in the sum of the five Equations). For the record, the solutions are $x_1 = 2,\ x_2 = 7,\ x_3 = 6,\ x_4 = 4$ and $x_5 = 3$.

Of course, if you have only two simultaneous Equations to solve, it is easy to write down explicit algebraic expressions for the solutions, and that may be the fastest and most efficient way of doing it. Thus, if

$$a_{11}x + a_{12}y = b_1 \tag{1.7.9}$$

and

$$a_{21}x + a_{22}y = b_2, \tag{1.7.10}$$

the solutions are

$$x = c(b_1a_{22} - b_2a_{12}) \tag{1.7.11}$$

and

$$y = c(b_2a_{11} - b_1a_{21}), \tag{1.7.12}$$

where

$$c = 1/(a_{11}a_{22} - a_{12}a_{21}). \tag{1.7.18}$$

1.8: Simultaneous Linear Equations, N > n

Consider the following Equations

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + b_1 = 0 \tag{1.8.1}$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + b_2 = 0 \tag{1.8.2}$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 + b_3 = 0 \tag{1.8.3}$$

$$a_{41}x_1 + a_{42}x_2 + a_{43}x_3 + b_4 = 0 \tag{1.8.4}$$

$$a_{51}x_1 + a_{52}x_2 + a_{53}x_3 + b_5 = 0 \tag{1.8.5}$$

Here we have five Equations in only three unknowns, and there is no solution that will satisfy all five Equations exactly. We refer to these Equations as the *Equations of condition*. The problem is to find the set of values of x_1, x_2 and x_3 that, while not satisfying any one of the Equations exactly, will come closest to satisfying all of them with as small an error as possible. The problem was well stated by Carl Friedrich Gauss in his famous *Theoria Motus*. In 1801 Gauss was faced with the problem of calculating the orbit of the newly discovered minor planet Ceres. The problem was to calculate the six elements of the planetary orbit, and he was faced with solving more than six Equations for six unknowns. In the course of this, he invented the method of least squares. It is hardly possible to describe the nature of the problem more clearly than did Gauss himself:

"...as all our observations, on account of the imperfection of the instruments and the senses, are only approximations to the truth, an orbit based only on the six absolutely necessary data may still be liable to considerable errors. In order to diminish these as much as possible, and thus to reach the greatest precision attainable, no other method will be given except to accumulate the greatest number of the most perfect observations, and to adjust the elements, not so as to satisfy this or that set of observations with absolute exactness, but so as to agree with all in the best possible manner."

If we can find some set of values of x_1, x_2 and x_3 that satisfy our five Equations fairly closely, but without necessarily satisfying any one of them exactly, we shall find that, when these values are substituted into the left hand sides of the Equations, the right hand sides will not be exactly zero, but will be a small number known as the residual, R.

Thus:

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + b_1 = R_1 \tag{1.8.6}$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + b_2 = R_2 \tag{1.8.7}$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 + b_3 = R_3 \tag{1.8.8}$$

$$a_{41}x_1 + a_{42}x_2 + a_{43}x_3 + b_4 = R_4 \tag{1.8.9}$$

$$a_{51}x_1 + a_{52}x_2 + a_{53}x_3 + b_5 = R_5 \tag{1.8.10}$$

Gauss proposed a "best" set of values such that, when substituted in the Equations, gives rise to a set of residuals such that the sum of the squares of the residuals is least. (It would in principle be possible to find a set of solutions that minimized the sum of the absolute values of the residuals, rather than their squares. It turns out that the analysis and the calculation involved is a good deal more difficult than minimizing the sum of the squares, with no very obvious advantage.) Let S be the sum of the squares of the residuals for a given set of values of x_1, x_2 and x_3. If any one of the x-values is changed, S will change - unless S is a minimum, in which case the derivative of S with respect to each variable is zero. The three Equations

$$\frac{\partial S}{\partial x_1} = 0, \quad \frac{\partial S}{\partial x_2} = 0, \quad \frac{\partial S}{\partial x_3} = 0 \tag{1.8.11}$$

express the conditions that the sum of the squares of the residuals is least with respect to each of the variables, and these three Equations are called the *normal Equations*. If the reader will write out the value of S in full in terms of the variables x_1, x_2 and x_3, he or she will find, by differentiation of S with respect to x_1, x_3 and x_3 in turn, that the three normal Equations are

$$A_{11}x_1 + A_{12}x_2 + A_{13}x_3 + B_1 = 0 \tag{1.8.12}$$

$$A_{12}x_1 + A_{22}x_2 + A_{23}x_3 + B_2 = 0 \tag{1.8.13}$$

$$A_{13}x_1 + A_{23}x_2 + A_{33}x_3 + B_3 = 0 \tag{1.8.14}$$

where

$$A_{11}=\sum a_{i1}^2,\quad A_{12}=\sum a_{i1}a_{i2},\quad A_{13}=\sum a_{i1}a_{i3},\quad B_1=\sum a_{i1}b_i, \tag{1.8.15}$$

$$A_{22}=\sum a_{i2}^2,\quad A_{23}=\sum a_{i2}a_{i3},\quad B_2=\sum a_{i2}b_i, \tag{1.8.16}$$

$$A_{33}=\sum a_{i3}^2,\quad B_3=\sum a_{i3}b_i, \tag{1.8.17}$$

and where each sum is from $i=1$ to $i=5$.

These three normal Equations, when solved for the three unknowns x_1, x_2 and x_3, will give the three values that will result in the lowest sum of the squares of the residuals of the original five Equations of condition.

Let us look at a numerical example, in which we show the running checks that are made in order to detect mistakes as they are made. Suppose the Equations of condition to be solved are

$$7x_1-6x_2+8x_3-15=0 \quad -6$$

$$3x_1+5x_2-2x_3-27=0 \quad -21$$

$$2x_1-2x_2+7x_3-20=0 \quad -13$$

$$4x_1+2x_2-5x_3-2=0 \quad -1$$

$$9x_1-8x_2+7x_3-5=0 \quad 3$$

$$-108 \quad -69 \quad -71$$

The column of numbers to the right of the Equations is the sum of the coefficients (including the constant term). Let us call these numbers s_1, s_2, s_3, s_4, s_5.

The three numbers below the Equations are $\sum a_{i1}s_i, \quad \sum a_{i2}s_i, \quad \sum a_{i3}s_i$

Set up the normal Equations:

$$159x_1-95x_2+107x_3-279=0 \quad -108$$

$$-95x_1+133x_2-138x_3+31=0 \quad -69$$

$$107x_1-138x_2+191x_3-231=0 \quad -71$$

The column of numbers to the right of the normal Equations is the sum of the coefficients (including the constant term). These numbers are equal to the row of numbers below the Equations of condition, and serve as a check that we have correctly set up the normal Equations. The solutions to the normal Equations are

$$x_1=2.474 \quad x_2=5.397 \quad x_3=3.723$$

and these are the numbers that satisfy the Equations of condition such that the sum of the squares of the residuals is a minimum.

I am going to suggest here that you write a computer program, in the language of your choice, for finding the least squares solutions for N Equations in n unknowns. You are going to need such a program over and over again in the future – not least when you come to Section 1.12 of this chapter!.

1.9: Nonlinear Simultaneous Equations

We consider two simultaneous Equations of the form

$$f(x,\ y)=0, \tag{1.9.1}$$

$$g(x,\ y)=0 \tag{1.9.2}$$

in which the Equations are not linear.

As an example, let us solve the Equations

$$x^2=\frac{a}{b-\cos y} \tag{1.9.3}$$

$$x^3-x^2=\frac{a(y-\sin y\cos y)}{\sin^3 y}, \tag{1.9.4}$$

in which a and b are constants whose values are assumed given in any particular case.

This may seem like an artificially contrived pair of Equations, but in fact a pair of Equations like this does appear in orbital theory.

We suggest here two methods of solving the Equations.

In the first, we note that in fact x can be eliminated from the two Equations to yield a single Equation in y:

$$F(y)=aR^3-R^2-2SR-S^2=0, \tag{1.9.5}$$

where

$$R=1/(b-\cos y) \tag{1.9.5a}$$

and

$$S=(y-\sin y\cos y)/\sin^3 y. \tag{1.9.5b}$$

This can be solved by the usual Newton-Raphson method, which is repeated application of $y=y-F/F'$. The derivative of F with respect to y is

$$F'=3aR^2R'-2RR'-2(S'R+SR')-2SS' \tag{1.9.6}$$

where

$$R'=-\frac{\sin y}{(b-\cos y)^2} \tag{1.9.6a}$$

and

$$S'=\frac{\sin y(1-\cos 2y)-3\cos y(y-\frac{1}{2}\sin 2y)}{\sin^4 y} \tag{1.9.6b}$$

In spite of what might appear at first glance to be some quite complicated Equations, it will be found that the Newton-Raphson process, $y=y-F/F'$, is quite straightforward to program, although, for computational purposes, F and F' are better written as

$$F=-S^2+R(-2S+R(-1+aR)), \tag{1.9.7a}$$

and

$$F'=3aR^2R'-2(R+S)(R'+S') \tag{1.9.7b}$$

Let us look at a particular example, say with $a=36$ and $b=4$. We must of course, make a first guess. In the orbital application, described in Chapter 13, we suggest a first guess. In the present case, with $a=36$ and $b=4$, one way would be to plot graphs of Equations 1.9.3 and 1.9.4 and see where they intersect. We have done this in Figure 1.4, from which we see that y must be close to 0.6.

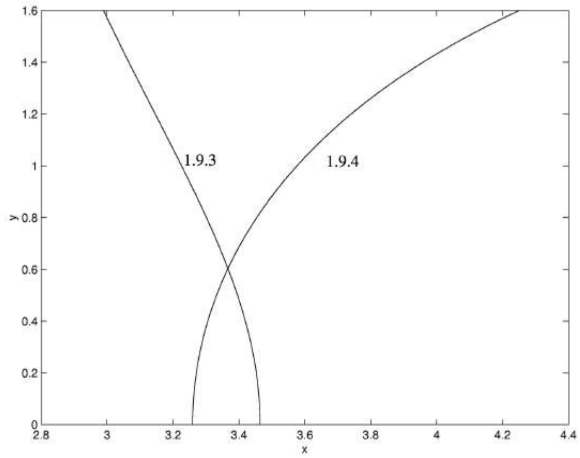


FIGURE 1.4
Equations 1.9.3 and 1.9.4 with $a = 36$ and $b = 4$.

With a first guess of $y = 0.6$, convergence to $y = 0.60292$ is reached in two iterations, and either of the two original Equations then gives $x = 3.3666$.

We were lucky in this case in that we found we were able to eliminate one of the variables and so reduce the problem to a single Equation on one unknown. However, there will be occasions where elimination of one of the unknowns may be considerably more difficult or, in the case of two simultaneous transcendental Equations, impossible by algebraic means. The following iterative method, an extension of the Newton-Raphson technique, can nearly always be used. We describe it for two Equations in two unknowns, but it can easily be extended to n Equations in n unknowns.

The Equations to be solved are

$$f(x,\ y) = 0 \tag{1.9.8}$$

$$g(x,\ y) = 0. \tag{1.9.9}$$

As with the solution of a single Equation, it is first necessary to guess at the solutions. This might be done in some cases by graphical methods. However, very often, as is common with the Newton-Raphson method, convergence is rapid even when the first guess is very wrong.

Suppose the initial guesses are $x + h$, $y + k$, where x, y are the correct solutions, and h and k are the errors of our guess. From a first-order Taylor expansion (or from common sense, if the Taylor expansion is forgotten),

$$f(x+h, y+k) \approx f(x,y) + hf_x + kf_y. \tag{1.9.10}$$

Here f_x and f_y are the partial derivatives and of course $f(x,\ y) = 0$. The same considerations apply to the second Equation, so we arrive at the two linear Equations in the errors h, k:

$$f_x h + f_y k = f, \tag{1.9.11}$$

$$g_x h + g_y k = g. \tag{1.9.12}$$

These can be solved for h and k:

$$h = \frac{g_y f - f_y g}{f_x g_y = f_y g_x}, \tag{1.9.13}$$

$$k = \frac{f_x g - g_x f}{f_x g_y - f_y g_x}. \tag{1.9.14}$$

These values of h and k are then subtracted from the first guess to obtain a better guess. The process is repeated until the changes in x and y are as small as desired for the particular application. It is easy to set up a computer program for solving any two Equations; all that will change from one pair of Equations to another are the definitions of the functions f and g and their partial derivatives.

In the case of our example, we have

$$f = x^2 - \frac{}{b - \cos y} \tag{1.9.15}$$

$$g = x^3 - x^2 - \frac{a(y - \sin y \cos y)}{\sin^3 y} \tag{1.9.16}$$

$$f_x = 2x \tag{1.9.17}$$

$$f_y = \frac{a \sin y}{(b - \cos y)^2} \tag{1.9.18}$$

$$g_x = x(3x - 2) \tag{1.9.19}$$

$$g_y = \frac{a[3(y - \sin y \cos y) \cos y - 2 \sin^3 y]}{\sin^4 y} \tag{1.9.20}$$

In the particular case where $a = 36$ and $b = 4$, we can start with a first guess (from the graph - Figure I.4) of $y = 0.6$ and hence $x = 3.3$. Convergence to one part in a million is reached in three iterations, the solutions being $x = 3.3666$, $y = 0.60292$.

A simple application of these considerations arises if you have to solve a polynomial Equation $f(z) = 0$, where there are no real roots, and all solutions for z are complex. You then merely write $z = x + iy$ and substitute this in the polynomial Equation. Then equate the real and imaginary parts separately, to obtain two Equations of the form

$$R(x,\ y) = 0 \tag{1.9.21}$$

$$I(x,\ y) = 0 \tag{1.9.22}$$

and solve them for x and y. For example, find the roots of the Equation

$$z^4 - 5z + 6 = 0. \tag{1.9.23}$$

It will soon be found that we have to solve

$$R(x,\ y) = x^4 - 6x^2y^2 + y^4 - 5x + 6 = 0 \tag{1.9.24}$$

$$I(x,\ y) = 4x^3 - 4xy^2 - 5 = 0 \tag{1.9.25}$$

It will have been observed that, in order to obtain the last Equation, we have divided through by y, which is permissible, since we know z to be complex. We also note that y now occurs only as y^2, so it will simplify things if we let $y^2 = Y$, and then solve the Equations

$$f(x, Y) = x^4 - 6x^2Y + Y^2 - 5x + 6 = 0 \tag{1.9.26}$$

$$g(x, Y) = 4x^3 - 4xY - 5 = 0 \tag{1.9.27}$$

It is then easy to solve either of these for Y as a function of x and hence to graph the two functions (figure I.5):

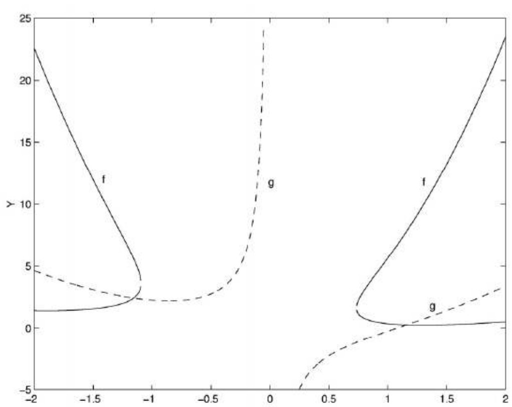


FIGURE I.5

This enables us to make a first guess for the solutions, namely

$$x = -1.2, \quad Y = 2.4$$

and

$$x=+1.2, \quad Y=0.25818$$

We can then refine the solutions by the extended Newton-Raphson technique to obtain

$$x=-1.15697, \quad Y=2.41899$$

$$x=+1.15697, \quad Y=0.25818$$

so the four solutions to the original Equation are

$$z=-1.15697 \pm 1.55531 i$$

$$z=1.15697 \pm 0.50812 i$$